(UIC & NU) Recycled polymer–PAL composites summary

Christian Fernandez · September 18, 2026

1 Objective

We want a data-driven model that links formulation and processing to the performance of extruded polymer–PAL composites, built on the mechanical data together with the rheology data, while also considering the recycle number. Every reprocessing loop degrades the material, so strength, stiffness, toughness and the flow curve are all expected to drift with the number of recycles. The degradation will be utilized during multi-objective optimization of the mechanical and rheological properties, and predicting how many times a formulation can be recycled before its performance is no longer usable. For the modeling part we are considering latent-variable Gaussian process for the categorical polymer input that also consider the heteroscedastic noise.

2 Inputs, outputs and design of experiments

Input	Type	Values / range
Polymer system	categorical	PP+PAL · HDPE+PAL · TPU+PAL
PAL content	continuous	10–50 %
Extrusion temperature	continuous	180–240 °C
Recycle number	categorical	1 / 5 / 10×, on every formulation

Output	Units	Test
UTS	MPa	tensile, ≥ 3 replicates
Young's modulus (stiffness)	MPa	tensile
Toughness	MJ/m^3	tensile
Strain at break	–	tensile
Viscosity / flow curve	Pa·s	rheometer, window 0.1–10 s^{-1}

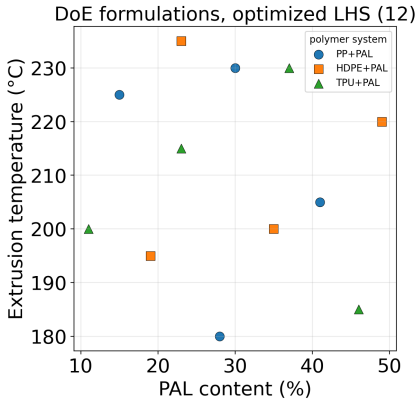


Figure 1: The 12 DoE formulations.

The formulation factors are sampled with an optimized Latin hypercube: **12 formulations**, four per polymer (Fig. 1). Every formulation is carried through the whole 1 / 5 / 10× recycle, giving **36 measurements** plus repeated samples.

3 First DoE data

The initial dataset contains only tensile measurements from 30 specimens, representing 9 of the 12 planned formulations. Both the PP+PAL and HDPE+PAL systems are fully sampled with 3 to 5 replicates each, whereas the TPU+PAL system currently includes only a single specimen; rheological data has not yet been collected. As shown in Figures 2 and 3, UTS ranges mainly from 6 to 14 MPa and Young's modulus from 350 to 810 MPa. Increasing the PAL content decreases both properties in PP and HDPE alike, with optimal performance observed at low PAL concentrations extruded at approximately 195–200°C. Notably, the single TPU specimen exhibits elastomer behavior, yielding a toughness of 2.2 MJ/m^3, which is roughly five times greater than any other formulation. Within a given polymer class, UTS and toughness are not competing objectives as it can be observed on Figure 4. The trade-offs exist between Young's modulus and toughness, and in the selection of the base polymer (such as TPU).

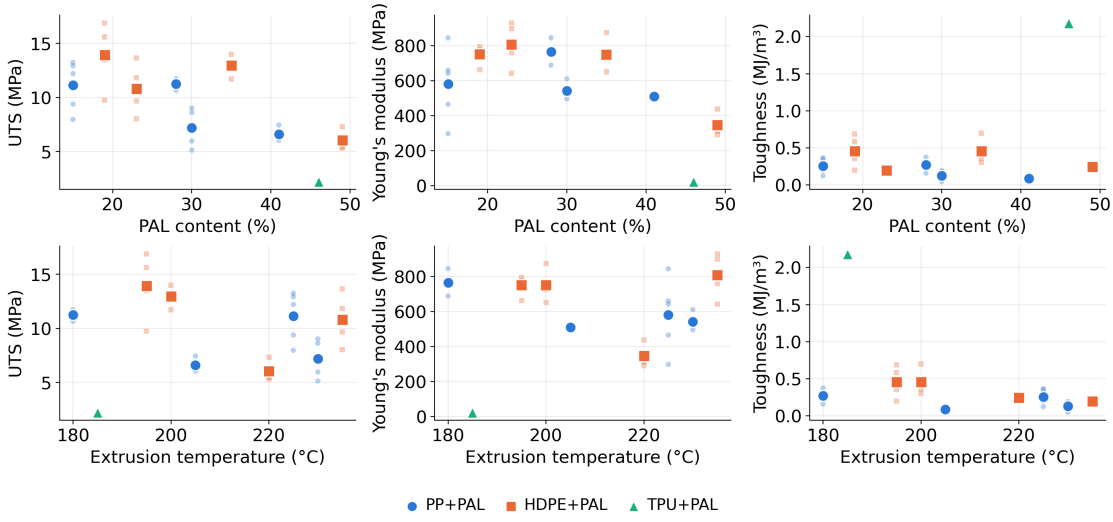


Figure 2: Tensile properties against the two continuous factors.

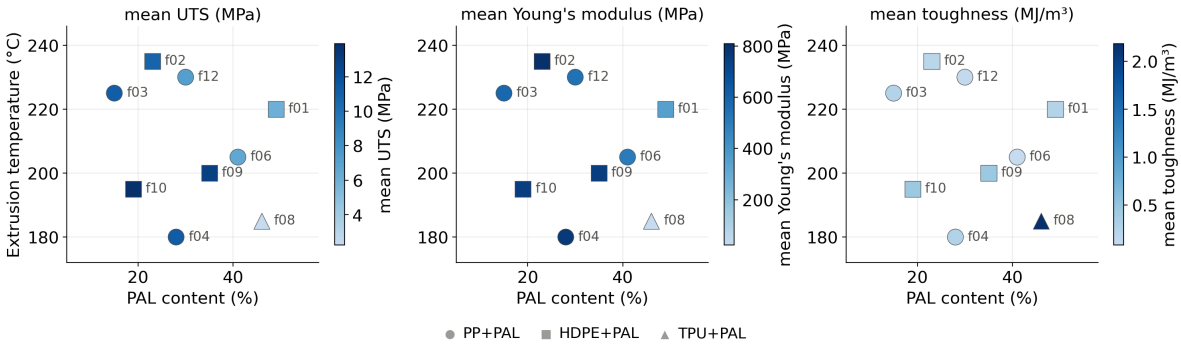


Figure 3: Formulation means placed in the design space.

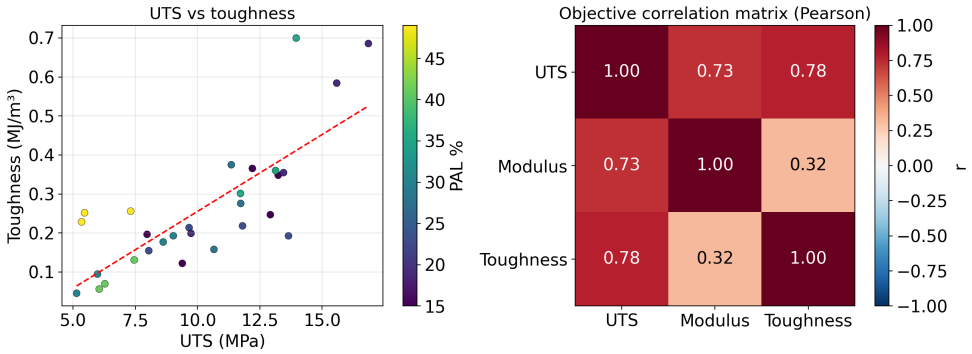


Figure 4: UTS vs. toughness per specimen, and the objective correlation matrix (PP+PAL and HDPE+PAL, $n = 29$).

4 Replicate noise

		PAL	T_{extr}		Young's modulus		UTS		Toughness	
Form.	Polymer	(%)	(°C)	n	mean (MPa)	CV (%)	mean (MPa)	CV (%)	mean (MJ/m^3)	CV (%)
f01	HDPE+PAL	49	220	3	347	22.9	6.0	18.3	0.25	6.0
f02	HDPE+PAL	23	235	4	809	16.4	10.8	22.7	0.19	14.9
f03	PP+PAL	15	225	5	583	35.7	11.1	21.0	0.26	40.0
f04	PP+PAL	28	180	3	767	10.2	11.2	4.9	0.27	40.3
f06	PP+PAL	41	205	3	510	3.6	6.6	11.4	0.09	46.3
f08	TPU+PAL	46	185	1	24	–	2.2	–	2.18	–
f09	HDPE+PAL	35	200	3	751	15.2	12.9	8.8	0.45	47.4
f10	HDPE+PAL	19	195	4	751	8.0	13.9	22.4	0.46	48.3
f12	PP+PAL	30	230	4	544	8.9	7.2	26.7	0.13	54.5

Table 1: Coefficient of variation (std/mean) across replicates, one shading scale for all CV columns; f08 has one specimen.

The experimental noise is significant and heteroscedastic, with coefficients of variation reaching 36 % for modulus, 27 % for UTS, and 55 % for toughness (Table 1). This variance changes across inputs and is likely affected by filament diameter variations (1.40–2.05 mm) and heterogeneous mixing among polymers.